

Features of a sample object

Every sample includes:

- Network topology
- Traffic parameters.
- Routing configuration.
- Slice reservations.
- Measurements (delay, jitter and loss, among others) and link performance from the resulting simulation

Only one path connecting each source-destination pair.

Simulation stops when the stationary state is detected. The simulation time to generate each sample is different.

Summary of global parameters:

- *global_packets*: Packets transmitted in the network per time unit (packets/time unit).
- *global_losses*: Packets lost in the network per time unit (packets/time unit).
- *global_delay*: Average per-packet delay over all the packets transmitted in the network (time units).
- *maxAvgLambda*: overall traffic intensity of the network scenario. Maximum average traffic rate (bits/time unit) that a path can generate in the simulation scenario. Note that this traffic rate may be split into several flows, sending traffic over the same src-dst path.
- *performance_matrix*: Matrix with aggregate src-dst and flow-level performance measurements (delay, jitter and loss) measured on each src-dst pair.
- *traffic_matrix*: Matrix with the time and size distributions used to generate traffic for each src-dst pair.
- *routing_matrix*: Matrix with the paths to connect every src-dst pair.
- *topology_object*: Networkx graph object including topology-related information at the node and link-level.
- *links_performance*: list of dictionaries with the performance metrics associated with each link. Not all datasets contain this information. In that case, this object is of type None.
- *port_stats*: list of dictionaries with the performance metrics associated with each output port. (see more details below)

performance_matrix: matrix that indexes performance measurements at the level of src-dst pairs. More than one flow can be exchanged on each src-dst pair. Two levels of granularity:

(i) for all aggregate flows on each src-dst pair

(ii) for every flow individually

Every element of this matrix (i.e., `performance_matrix[src,dst]`) contains a dictionary with:

- 'AggInfo': dictionary with performance measurements for all aggregate flows between a specific [src,dst] pair.
 - 'PktsDrop': packets dropped per time unit over the path [src,dst]. (packets/time unit)
 - 'AvgDelay': average per-packet delay over the packets transmitted on path [src,dst] (time units)
 - 'AvgLnDelay': average of the neperian logarithm of the per-packet delay over all packets transmitted on path [src, dst]. This is "avg(ln(packet_delay))"
 - 'p10', 'p20', 'p50', 'p80', 'p90': percentiles (10, 20, 50, 80, and 90) of the per-packet delay distribution on path [src,dst]. (time units)
 - 'Jitter': Variance of the per-packet delay over the packets transmitted on path [src,dst]. This is var(packet_delay)
- 'Flows': List of dictionaries with performance measurements for each flow between node i and node j. Flows on each src-dst pair are indexed by a numerical ID (e.g., `performance_matrix[0,1]['Flows']['0']`). Each flow contains the following keys:
 - 'PktsDrop': packets dropped per time unit on the flow. (packets/time unit)
 - 'AvgDelay': average per-packet delay on the flow. (time units)
 - 'AvgLnDelay': average of the neperian logarithm of the per-packet delay over all packets transmitted on the flow. i.e., "avg(ln(packet_delay))"
 - 'p10', 'p20', 'p50', 'p80', 'p90': percentiles (10, 20, 50, 80, and 90) of the per-packet delay distribution on the flow. (time units)
 - 'Jitter': variance of the per-packet delay (i.e., jitter) on the flow. i.e., var(packet_delay)

Access to the information as follows:

- `perf[src,dst]`: dictionary with the performance measurements for the communication between node src and node dst.
- `perf[src,dst]['AggInfo']`: dictionary with performance metrics on aggregate traffic between node src and node dst. Use `perf[src,dst]['AggInfo']['<nameparam>']`.
- `perf[src,dst]['Flows']`: flow-level performance metrics on traffic between node src and node dst. Use `perf[src,dst]['Flows']['<numflow>']['<nameparam>']`

traffic_matrix: traffic-related information at the level of src-dst pairs. Also two levels of granularity:

(i) for all aggregate flows on each src-dst pair

(ii) for every flow individually

Every element of this matrix (i.e., `traffic_matrix[src,dst]`) contains a dictionary with:

- 'AggInfo': dictionary with traffic measurements for all aggregate flows between a specific [src,dst] pair.
 - 'AvgBw': average bandwidth from node src to node dst (bits/time unit).
 - 'PktsGen': packets generated from node src to node dst per time unit (packets/time unit).
- 'Flows': List of dictionaries with flow-level traffic information for each flow between node src and node dst. It includes the traffic measurements for each flow and the parameters used to generate the traffic flow in the network simulator.
 - 'TimeDis': inter-packet arrival time distribution used to generate the traffic of the flow.
 - 'TimeDistParams': dictionary with the parameters of the specific inter-packet arrival time distribution.
 - 'EqLambda': average bitrate per time unit (bits/time unit).
 - 'AvgPktsLambda': average number of packets of average size generated per time unit (packets/time unit).
 - 'MinPktLambda': minimum number of packets generated per time unit (packets/time unit).
 - 'MaxPktLambda': maximum number of packets generated per time unit (packets/time unit).
 - 'StdDev': Standard deviation of the interarrival time (used in the normal distribution).
 - 'PktsLambdaOn': Average packets per time unit generated during the ON period (packets/time unit).
 - 'AvgTOff': Average duration of the OFF period. The duration of every OFF period is modeled with an exponential distribution of average = 'AvgTOff'. (time units)
 - 'AvgTON': Average duration of the ON period. The duration of every ON period is modeled with an exponential distribution of average = 'AvgTON'. (time units)
 - 'ExpMaxFactor': Factor used to define an upper bound for exponential distributions. The upper bound is defined as: 'ExpMaxFactor'* lambda of the exponential
 - 'BurstGenLambda': Burst generation rate of the flow (bursts/time unit).
 - 'Bitrate': Bitrate during a burst: (bits/time unit).
 - 'ParetoMinSize': Minimum number of bits in the burst (bits).
 - 'ParetoMaxSize': Maximum number of bits in the burst (bits).
 - 'ParetoAlfa': Shape parameter (α) used in the Pareto distribution.
 - 'SizeDist': packet size distribution used for the flow (see possible packet size distributions in Section 4.2).
 - 'SizeDistParams': dictionary with the parameters of the specific size distribution (see the parameters of each packet size distribution in Section 4.2).
 - 'AvgPktSize': Average packet size (bits).
 - 'MinSize': Minimum packet size (bits).
 - 'MaxSize': Maximum packet size (bits).
 - 'PktSize1': First packet size option (bits).
 - 'PktSize2': Second packet size option (bits).
 - 'NumCandidates': Number of different packet sizes considered.
 - 'Size_i': Size of the candidate packet i (bits).

- 'Prob_i': Probability to select candidate packet i (bits).
- 'AvgBw': average bandwidth for this flow (bits/time unit).
- 'PktsGen': packets generated by this flow per time unit (packets/time unit).
- 'ToS': Type of Service associated with the flow defined as an integer.

In case of an exponential inter-packet time distribution, the parameters of the first flow can be accessed using the following lines of code:

```
traffic_matrix[src,dst]['Flows']['0']['TimeDistParams']['EqLambda']
traffic_matrix[src,dst]['Flows']['0']['TimeDistParams']['AvgPktsLambda']
traffic_matrix[src,dst]['Flows']['0']['TimeDistParams']['ExpMaxFactor']
```

Note: If the dataset only has one flow per path, then traffic and performance measurements over the aggregate traffic on a src-dst pair are the same as flow-level measurements. This applies to performance_matrix and traffic_matrix. Using information at the flow-level is recommended for the traffic matrix (e.g., traffic_matrix[0,1]['Flows']['0']), since it includes additional information not considered at the level of aggregate traffic.

routing_matrix: includes all the paths connecting every src-dst pair. "route[src,dst]" returns a list describing the path from node src to node dst. This list includes the IDs of the nodes that the path traverses. All the flows of a src-dst pair follow the same path.

topology_object: Graph object from the Networkx library Information on nodes and links (i.e., edges). Access the data as follows:

- g.nodes: Returns a list with all the node IDs.
- g.nodes[id]: Returns a dictionary with all the information parameters of the selected node.
- g.edges: Returns a list of tuples describing the topology edges. Each tuple is described as (src node ID, dst node ID, link ID). Link ID is always '0' as only one link for the same src-dst pair is supported.
- g[src][dst][0]: Dictionary with the information parameters of the (directed) link between src and dst.

port_stats: List of nested dictionaries. Contains the performance metrics of the output node ports. The outer list contains a dictionary of dictionaries for each node. The first dictionary contains the list of adjacent nodes and the last dictionary contains the statistics of the outgoing port used to reach the adjacent node. The performance dictionary contains the following keys:

- 'utilization': Average utilization of the outgoing port (in the range [0, 1]).
- 'losses': Average packets lost in the outgoing port (in the range [0, 1]).
- 'avgPacketSize': Average packet size from all outgoing packets going through the port (bits).
- 'qosQueuesStats': List of dictionaries containing the statistics for each QoS queue.
 - 'utilization': Average utilization of the outgoing port corresponding to the traffic associate with the QoS queue (in the range [0, 1])
 - 'losses': Average packets lost in the outgoing QoS queue (in the range [0, 1]).
 - 'avgPacketSize': Average packet size from all outgoing packets going through the QoS queue (bits).
 - 'avgPortOccupancy': Average port occupancy (service and waiting queue) of the QoS queue (packets).
 - 'maxQueueOccupancy': Maximum occupancy seen on the QoS queue.

Parameters of time distributions

- EXPONENTIAL_T: EqLambda, AvgPktsLambda, ExpMaxFactor
- DETERMINISTIC_T: EqLambda, AvgPktsLambda
- UNIFORM_T: EqLambda, MinPktLambda, MaxPktLambda
- NORMAL_T: EqLambda, AvgPktsLambda, StdDev
- ONOFF_T: EqLambda, PktsLambdaOn, AvgTOff, AvgTON, ExpMaxFactor
- PPBP_T: EqLambda, BurstGenLambda, Bitrate, ParetoMinSize, ParetoMaxSize, ParetoAlfa, ExpMaxFactor

Parameters of packet size distributions

- DETERMINISTIC_S: AvgPktSize
- UNIFORM_S: AvgPktSize, MinSize, MaxSize
- BINOMIAL_S: AvgPktSize, PktSize1, PktSize2
- GENERIC_S: AvgPktSize, NumCandidates, Size_i, Prob_i

List of node parameters

- 'levelsQoS': Number of supported QoS classes
- 'queueSizes': queue sizes (in number of packets) for each output queue of the node. Values are separated by commas. Each value is associated with a QoS queue.
- 'schedulingPolicy': 'FIFO', 'SP' (Strict Priority), 'WFQ' (Weighted Fair Queuing), 'DRR' (Deficit Round Robin). 'FIFO' policy is considered by default.
- 'schedulingWeights': List of weights separated by commas associated to each QoS queue. Used only in the WFQ and DRR.

List of link parameters

- 'port': Port number of the source node that links with the destination node.
- 'bandwidth': Link bandwidth (bits/time unit).
- 'weight': Not used.

List of slice parameters

Each sample includes a list of dictionaries, including slice data. Each dictionary in the list includes:

- origin_node: integer. id of node generating traffic for a flow in a slice.
- destination: integer. id of node receiving traffic for a flow in a slice.
- origin_node_antenna: integer. Next node (second) after the origin_node. Represents the antenna.
- path: list. List of hops of a flow, including origin and destination nodes.
- bandwidth: integer. traffic in bits per time unit of the flow.
- traffic_string: string. Data already included in the traffic matrix. This is the traffic string used in the simulator. Includes type of traffic (Poisson or CBR), bandwidth (Eq. Lambda), avg packets lambda and exponential max factor (only for Poisson). Also includes the packets size distribution: size distribution, average packet size and minimum/maximum packet sizes only for binomial distribution.
- delta: float. delta for the entire slice (all flows of a slice share the same value).
- type: string. "eMBB", "mMTC" or "URLLC"
- number: integer. slice identifier.